

2018556053 Cahit Barkın Özer

Subject :

Date :/...../.....

1 - a) Select S.SNumber, S.SName FROM Students S, Enrolled E
WHERE S.SNumber = E.SNumber AND
E.CourseID IN (SELECT E2.CourseID FROM Student S2,
Enroll E2 WHERE E2.SNumber LIKE
"Spring-2021 Semester" GROUP BY E.CourseID
HAVING COUNT(*) > 4

b) SELECT SName FROM Students Where Grade = 100

c) SELECT MAX(Grade) FROM Students S Where

d) SELECT DISTINCT T.TName FROM Course, Teacher T,
Enrolled E WHERE T.CourseID = E.CourseID AND E.CourseID =
T.CourseID AND E.Semester = "Spring-2021 Semester"

$$\left\lceil 1.93, \left\lceil \frac{200}{1.1} \right\rceil \right\rceil + 1 = 3$$

$$\text{Cost? } \text{Solving}(\text{cost}(M)) + \text{Solving}(\text{cost}(N)) + (M+N)$$

2 - a) Cost of Solving: $2M \times (\text{Number of passes})$

$$\left\lceil 1.93, \left\lceil \frac{2000}{1.1} \right\rceil \right\rceil \cdot 2 = 4$$

$$\text{Number of passes} = \left\lceil 1.93 \cdot \left\lceil \frac{M}{B} \right\rceil + 1 \right\rceil$$

$$\text{Cost} = 2 \cdot 3 \cdot 200 + 4 \cdot 2 \cdot 2000 + 2200 =$$

$$\boxed{19400}$$

SELECT S.SNumber, E.SNumber

b - FROM Students S, Enrolled E

WHERE E.SNumber = S.SNumber

AND E.Grade > 70 AND E.CourseID = "CEN350"

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3 Primary keys = { SNumber, CourseID, TNumber, SNumberCourseID Semester }

TNumber $\rightarrow$ CourseID

SNumber $\rightarrow$ SName

TNumber $\rightarrow$ TName

SNumber, CourseID, Semester $\rightarrow$ Grade

4 a) Unrepeatable read problem : T1 reads B, then T2 writes B, after T2 writes A again. The value read by T1 has been changed. Lost Write Problem : T2 writes B, T1 writes B again. T2 writes B.

b) T1: S(B)R(B) T2: S(B)R(B)X(A)W(A)C
T1: S(B)R(B)X(B)W(B)R(A)C

T2 has to wait for T1 to complete

Continues

T1: X(B)W(B)X

T2: X(B)W(B)X(A)W(A)C